PROJECT OVERVIEW

SignalOps

Multi-tenant emergency notification and employee-safety communication platform.

What SignalOps does
It gives organisations one controlled system to prepare, approve, deliver and audit emergency messages while tracking employee safety responses across facilities.

Product and technical context

September 2026

PURPOSE

A single operational channel for critical communication

SignalOps is designed for organisations that must notify people quickly, prove that messages were delivered, and understand who is safe during an incident.

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Communicate

Send prepared or incident-specific alerts to an entire organisation, location, building, group or individual.

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Account for people

Track safe acknowledgements, non-response and assistance requests against an immutable recipient snapshot.

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Govern

Use permissions, approval policies and audit history to control who can create, approve, release and resolve alerts.

PRIMARY USERS

Organisation administrators and controllers

- Maintain employees and operational locations
- Prepare emergency message templates
- Create and release alerts
- Approve sensitive alerts when policy requires it
- Monitor delivery and employee responses

RECIPIENTS

Employees using the mobile experience

- Receive alerts relevant to their assignment
- Review incident instructions
- Confirm that they are safe
- Request assistance with an optional note
- Access facility and muster-point context

Commercial model: each customer is represented as an isolated organisation/tenant. The same platform can serve multiple customers without mixing their users, facilities, alerts or audit records.

OPERATIONAL LIFECYCLE

From prepared message to accountable response

Prepare		Govern		Deliver		Respond
Templates, audience, channels and acknowledgement policy	→	Draft, submit, approve, return or emergency release	→	Email and Android push through an outbox worker	→	Safe, awaiting response or assistance required

STAGE	MEANING	CONTROL
Draft	The alert and recipient snapshot exist, but no delivery has been released.	Creator can submit; authorised sender can release.
Awaiting approval	A second authorised person must review the alert.	Self-approval is prohibited.
Active	The alert has been released and appears in operational monitoring.	Delivery, acknowledgements and assistance are tracked.
Resolved / cancelled	The incident lifecycle has ended while its records remain available.	History and audit evidence are retained.

Audience model

An alert can target the whole organisation, a facility, building, department, saved group or individual employee. Recipients are resolved and snapshotted when the alert is created.

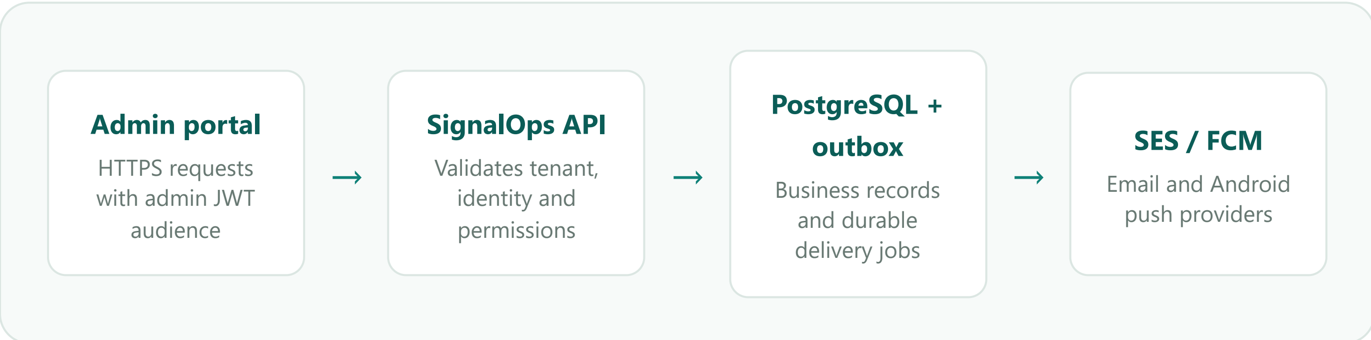
Delivery evidence

Summary metrics count unique people reached. Alert detail retains channel-level delivery status, provider information, timestamps and failure codes for every recipient.

APPLICATION ARCHITECTURE

Three applications connected by one tenant-scoped API

COMPONENT	RESPONSIBILITY	TECHNOLOGY
Admin portal	Organisation directory, facilities, templates, roles, alert lifecycle, delivery detail, responses and audit.	React 19, TypeScript, Vite, Netlify
API	Authentication, validation, permissions, tenant isolation, domain workflows and read models.	Node.js 22, Express 5, Zod
Delivery worker	Processes durable outbox jobs, invokes providers, records attempts and retries failures.	Node.js, PostgreSQL outbox
Employee app	Recipient alert feed, safety acknowledgement, assistance and device registration.	Flutter / Android
Data store	Tenants, users, assignments, alerts, recipients, delivery attempts, responses and append-only audit events.	PostgreSQL on AWS RDS



The employee application communicates back through the same API using a separate mobile JWT audience. Acknowledgements, assistance requests and device events are always verified against the recipient's tenant and user identity.

DELIVERY DESIGN

Transactional outbox rather than direct provider calls

Releasing an alert writes delivery records and outbox jobs inside the database transaction. A separate worker claims and processes those jobs. This avoids losing notifications if a provider is slow or unavailable and creates a durable history of retries and failures.

CORE INFORMATION MODEL

Organisation structure and incident evidence

Organisation directory

- Tenant and organisation settings
- Portal users and employees
- Departments and job assignments
- Facilities, buildings and muster points
- Saved recipient groups
- Roles and explicit permissions

Alert and response records

- Templates and delivery channels
- Alerts and lifecycle timestamps
- Audience definitions and recipient snapshots
- Per-recipient delivery attempts
- Acknowledgements and assistance requests
- Approvals and immutable audit events

CHANNEL MODEL

CHANNEL	ROLE IN SIGNALOPS	IMPLEMENTATION
Email	Operational alert delivery, invitations and password recovery.	AWS SES using the EC2 instance role.
Android push	Time-sensitive delivery to registered employee devices.	Firebase Cloud Messaging provider adapter.
SMS	Represented in the domain for future extension.	No provider is enabled; the product does not present it as operational.

AUTHENTICATION MODEL

Portal identity

Admin access tokens use the `signalops-admin` audience.

Mobile identity

Employee access tokens use the `signalops-mobile` audience.

Session safety

Short-lived access tokens and rotating, hashed refresh tokens protect long-running sessions.

TENANT AND ACCESS BOUNDARIES

Isolation and least privilege are part of every request

Tenant isolation

Business endpoints take tenant identity from the verified token, not from a client-supplied tenant identifier. Queries and mutations scope records using that trusted tenant context.

Permission enforcement

Backend permissions are authoritative. The portal mirrors them by hiding or disabling actions the current role cannot execute.

Approval separation

A creator cannot approve their own approval-required alert. A different authorised person must perform the review.

Audit integrity

Operational audit records are append-only at the database level and include actor, request, entity, timestamps, and before/after context.

> **Secret handling:** provider credentials stay in environment configuration, the EC2 instance role or a managed secret store. Tenant channel settings contain non-secret preferences only.

ACCOUNT SECURITY

- Passwords are stored using one-way hashing.
- Invitations use single-use activation tokens.
- Password recovery uses a six-digit email OTP with expiration and keyed verification.
- Forgot-password responses do not reveal whether an address exists.
- Successful password reset invalidates existing refresh sessions.

PRODUCTION TOPOLOGY

Isolated services on shared AWS infrastructure

SignalOps shares an AWS account, EC2 instance and RDS instance with another product, but its runtime and data remain isolated.

LAYER	SIGNALOPS CONFIGURATION
Admin portal	`https://signal-ops.netlify.app/`
Public API	`https://signalops-api.iot-cspllabs.com/api/v1`
Internal API listener	`127.0.0.1:5002` behind Nginx
Process manager	`signalops-api` and `signalops-worker` under PM2
Database	Dedicated SignalOps PostgreSQL database and least-privilege application role on RDS
Email identity	Verified SES domain `iot-cspllabs.com`; operational sender `notifications@iot-cspllabs.com`
AWS region	`ap-south-1`

SOURCE LAYOUT

APPLICATION	LOCAL FOLDER	REPOSITORY
Admin frontend	`D:\Notification_platform`	`amey990/signalops-frontend`
Backend	`D:\Notification-platform-backend`	`commedia-solutions/Notification-platform-backend`
Employee app	`D:\Notification-platform-app`	Flutter application source

TECHNOLOGY SUMMARY

React · TypeScript · Vite · Flutter · Node.js · Express · PostgreSQL · AWS RDS · AWS SES · Firebase FCM · PM2 · Nginx · Netlify · JWT · Zod

IMPLEMENTATION INTENT

Operational truth over demonstration-only behaviour

Real data

Counts, statuses and assignments must come from backend records. The product avoids fake connectivity, safety or occupancy states.

Traceable actions

Important state changes create durable records so an organisation can reconstruct what happened and who acted.

Least privilege

Users see and perform only the operations granted by their organisation role.

Preparedness with flexibility

Templates accelerate emergency communication while controlled editing supports the facts of a live incident.

Reliable delivery

Provider calls are asynchronous and retryable; delivery evidence is stored separately from alert business state.

Customer isolation

Multi-tenant design is retained internally even when the user interface presents a single organisation context.

CURRENT CUSTOMER CONTEXT

The initial organisation is **Avenir Manufacturing**. The portal is structured as that organisation’s command centre while the backend remains suitable for additional customer tenants.